Apertus Hackathon, Track 2A (OST): Multilingual Natural Language Inference over Swiss Official Voting Booklets

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1 Summary

Every Swiss popular vote comes with an official booklet in German, French and Italian (and Romansh, not included in this study). The task is to decide whether the booklet supports a claim about the vote (entailment), contradicts it, or neither (neutral). Two thirds of the claims are in another language than the booklet. The beginner task gives a section of the booklet, the advanced task the whole PDF, and asks for the evidence too. Both our solutions retrieve, then verify [18]. Our code finds the few sentences that bear on the claim, and `Apertus-v1.5-70B` judges only those. On all 1,488 public cases, we reach a macro-F1 of 0.987 in the beginner task (baseline 0.788), with 834 tokens and 0.6 ms of processing time per case. In the advanced task we reach 0.974, with an evidence Hit@5 of 0.979, 1,157 tokens and 3.6 ms per case. Prompting with the whole booklet reached 0.70–0.73, for about 34,770 tokens per case.

2 Architecture

Component	What it does	Where it runs
The CLI, `src/entrypoint.py`	Reads the cases, decides 8 at a time, writes one JSON line per case	Our Docker image: linux/amd64, CPU only
PDFium (`pypdfium2`)	Reads the booklet's text from its PDF tags	Our Docker image
BM25, our own Python code	Finds the sentences that bear on the claim	Our Docker image
Apertus v1.5 70B	Translates a claim in another language, and decides the label	A hosted endpoint, at `BASE_URL`

There is no OCR, no embedding model and no other model. A case sends one request to Apertus, or two if the claim is in another language than the document, a translation, then the decision. In Figures 1 and 2, the numbers are the steps of Sections 2.2 and 2.3, and a dashed box is a step that only some cases take.

2.1 Shared by both tasks: sentence search and translation

- **Sentences** are the unit of search, since the evidence has to be short and verbatim. Chen et al. (2024) [3] found, with dense retrievers, that sentences fit more of the relevant text into a fixed number of tokens than passages, and propositions more still. But propositions are rewritten by a model, so they could not be quoted. A sentence never crosses a paragraph or a page, an abbreviation such as "z. B." does not end it, and one under 40 characters joins the next.
- **BM25** ranks the sentences for the claim ($k_1 = 1.2$, $b = 0.75$; Robertson and Zaragoza, 2009 [16]). Words are lower-cased, stripped of accents and cut to their first 7 letters. This is a language-independent stand-in for stemming (McNamee et al., 2009 [16]), which found more evidence than 5, 6 or whole words. A sentence also gets 30 % of its better neighbour's score.
- **Translation:** BM25 needs the claim in the document's language, so Apertus first translates a claim in another language. This is translate-test (Conneau et al., 2018 [4]; Artetxe et al., 2023 [1]), with the model translating for itself (self-translate; Etxaniz et al., 2024 [7]).

2.2 Beginner task: a scoped reference

1. **Translate** a claim into the reference's language if it is not the same.
2. **Scope** a reference over 2,000 characters, as 1,225 of the 1,488 are: the 6 sentences that match the claim best, each with the sentence before and after it, in order, with "[. . .]" for each gap.
3. **Decide:** Apertus explains in at most three sentences what the reference says about the claim, then writes `Label: <word>` (Section 3).

The evidence is the reference shown, `none` evidence and page `null` for a neutral answer.

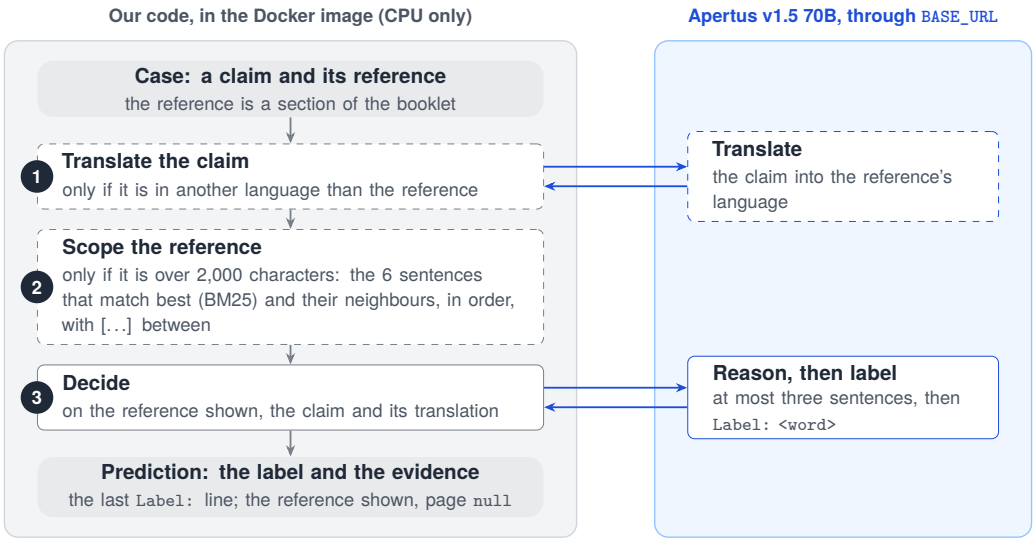


Figure 1: Beginner task

2.3 Advanced task: retrieve, then verify

1. **Read the booklet** once, when a case first needs it, from its PDF tags (ISO 32000-2 §14.8; PDF/UA, ISO 14289-1): a structure tree of headings and paragraphs in reading order, which leaves out running headers and page numbers. The text layer stands in where tags are missing, for the untagged voting text, and for the one booklet without tags (French, 28 November 2021).
2. **Index its sentences,** each with its page and the two headings above it.
3. **Narrow to the claim's vote.** Running headers such as "Erste Vorlage: ..." assign the pages to the votes, and we keep the vote whose pages share the most rare words and word pairs with the case's vote title. That is a third of a booklet on average, with all gold pages in 989 of 990 cases.
4. **Search** these pages with the claim, or with its translation.
5. **Build the excerpts:** the 8 best sentences, each with its neighbours on the page, about 2,000 characters, numbered and labelled with page and section, e.g. `[3] (p. 12, Argumente › Zu strikte Vorgaben)`.
6. **Decide** as in the beginner task, and name the deciding excerpts: `Excerpts: <numbers>`.
7. **Return the evidence:** five verbatim sentences with their pages, those of the cited excerpts first, but never the vote's title, which starts every gold passage. A neutral answer gets evidence too, since the evidence of every gold entailment or contradiction is scored.

Choice	Why
About 2,000 characters of context	References have up to 25,000 characters, booklets 23,000 to 270,000. Long and distracting contexts lower accuracy [11, 17], and short retrieved contexts can match long-context models [20].
Only the claim's vote	Claims about the voting text or the summary read alike for every vote, so they had often been judged against another vote's text.
The PDF tags	They kept 97.9 % of each gold passage, at 3 ms per page; the text layer 96.0 % at 1 ms, and Docling [2] 95.9 % at 157 ms.
Only the supplied booklet	The challenge asks to judge against the supplied source. Also searching its edition in the claim's language raised macro-F1 from 0.951 to 0.962, but that edition is in the input only when another case uses it.
Neutral first, contradiction only with a quotable sentence	Most of the baseline's errors were neutral claims answered contradiction (Section 5). As in answers backed by quotes [12], a contradiction has to rest on a sentence the model can quote; the quote itself is not checked.
Reasoning before the label	Reasoning first helps [19], and an explanation written after the label cannot change it.

3 Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-70B` [9], inference only, prompted zero-shot at temperature 0. The 8B model served during development and scored lower.
- **Calls:** the translation of a cross-lingual claim (at most 200 output tokens), and the decision: at most three sentences of reasoning, then the label (at most 200 output tokens; 250 in the advanced task, whose answer

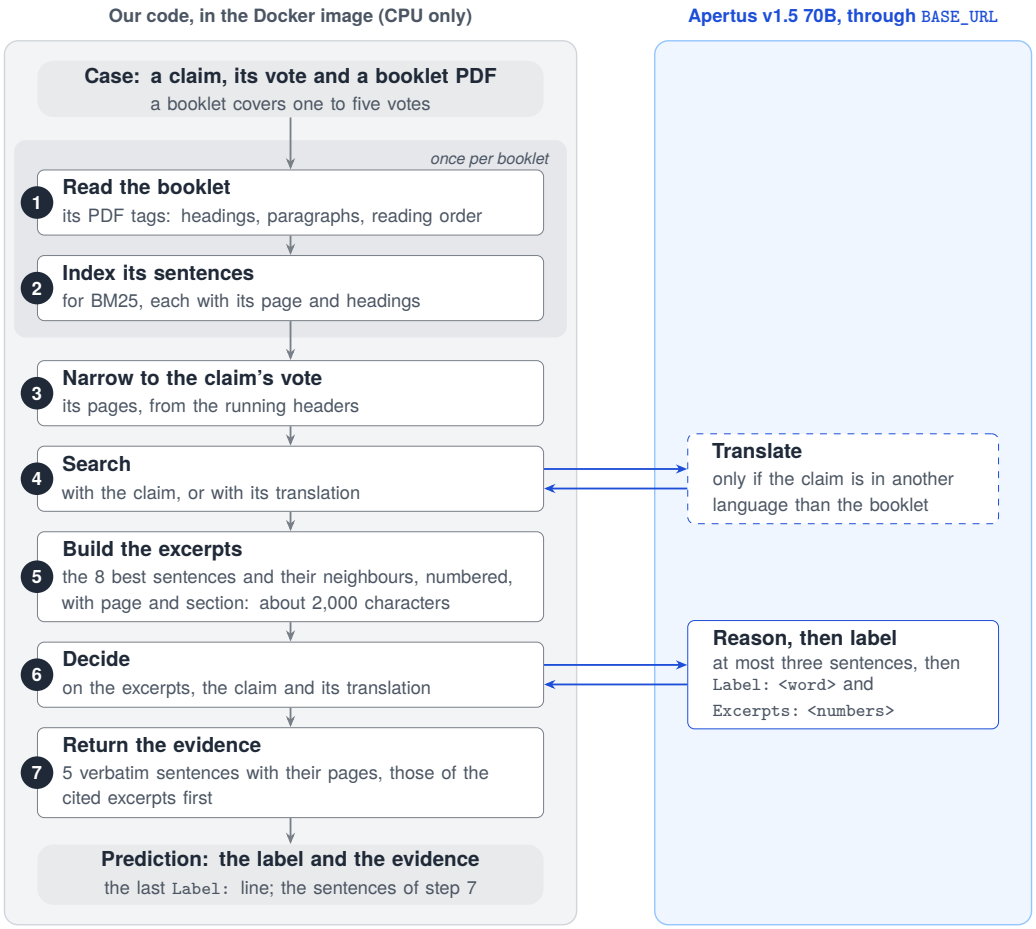


Figure 2: Advanced task

also names the excerpts). Only the last `Label:` and `Excerpts:` lines are read, since the reasoning may name other labels and numbers.

- **Endpoint:** the OpenAI-compatible chat API at `BASE_URL`, the CSCS Inference API during development and the organisers' proxy at the evaluation, with 8 requests in parallel.
- **Prompt:** the label definitions below, then the claim with its translation. In the advanced task it says "booklet" for "reference", gives the vote and the numbered excerpts, and says that only the booklet counts, not the model's own knowledge.

  neutral: the reference neither supports nor contradicts the claim
  contradiction: the reference says the opposite of the claim, and you can quote the sentence that does
  entailment: the reference supports the claim
  If the reference does not mention what the claim says, or is about another topic, the label is "neutral", not "contradiction": a claim the reference does not support is not contradicted by it.

4 Data

- **Dataset:** `OSTswiss/MNLIoverSwissVotingBooklets`, version v1.1, pinned to one revision [14], MIT licence. Its 1,488 rows each give a beginner and an advanced case with the same claim and label. The labels are balanced, the claims split evenly between German, French and Italian, and two thirds of the cases are cross-lingual.
- **Booklets:** the 60 official booklets the dataset refers to, from the Federal Chancellery (`bk.admin.ch`). They are public and not redistributed, except one for the `make run` sample. There is no personal data. Swiss voting booklets have also been studied for stance detection [6].

5 Evaluation

We report the challenge's metrics on all 1,488 public cases: macro-F1; evidence Hit@5, whether one of the first five evidence items matches the gold passage (rapidfuzz `partial_ratio` ≥ 90, as in the challenge's `evaluate.py`); tokens per case, as the API reports them; and processing time per case, without Apertus' latency (`src/timing.py`). All runs used Apertus v1.5 70B on the CSCS Inference API. Repeated identical requests changed 4 of 515 labels, so differences under about 0.005 are noise.

Table 1: Results (all 1,488 cases, unless noted; ↑ higher is better, ↓ lower is better). Time is the processing time per case, without Apertus' latency.

	Macro-F1↑	Hit@5↑	Tokens per case (in + out)↓	Time↓
Beginner task				
Baseline: the whole reference, a one-word answer	0.788	–	1,897 + ≤ 10	–
Our submitted model	**0.987**	–	748 + 86	0.6 ms
Advanced task				
Baseline: full-document prompting, the whole booklet (150 cases, 2 runs)	0.701 / 0.727	0.128	34,762 + ≤ 10	–
Ours, on the same 150 cases	0.978	1.000	1,048 + 102	–
Our submitted model	**0.974**	**0.979**	1,053 + 104	3.6 ms

Table 2: What each step added (all 1,488 cases, unless noted)

Step	Macro-F1↑	Hit@5↑	Input tokens per case↓
Beginner task			
Baseline prompt	0.788	–	1,897
Neutral first, reasoning before the label	0.901	–	1,952
`quote` definitions	0.962	–	1,984
+ the translation of a cross-lingual claim	0.965	–	2,092
+ only the sentences around the claim (ours)	**0.987**	–	**748**
Advanced task			
Full-document prompting (150 cases)	0.70–0.73	0.13	34,762
Only the vote's pages; the model's quotes are looked up in the PDF	0.850	0.485	11,553
Retrieve, then verify: BM25 over the vote's pages with the translated claim, the label first	0.951	0.978	1,077
+ reasoning before the label (ours)	**0.974**	b	**1,053**

Languages. Cross-lingual cases do as well as same-language ones. Macro-F1 0.985 against 0.990 in the beginner task, and 0.982 against 0.958 in the advanced task. By the language of the claim or of the document, macro-F1 stays between 0.980 and 0.990 (beginner) and between 0.963 and 0.987 (advanced). The weakest pair is German claims on German booklets, 0.923 in the advanced task.

What made the difference was the line between neutral and contradiction. In the beginner task, 274 of the baseline's 498 errors were neutral claims answered contradiction; putting neutral first with reasoning cut them to 140, and the `quote` definitions to 46. Scoping fixed 44 of the 52 errors left, since irrelevant context distracts language models [17], and made 12 new ones. In the advanced task, answering with the label first missed 42 contradictions, also with 250 tokens instead of 40. Reasoning first left 4.

Tried and left out.

- **Verified quotes,** our first advanced approach: Apertus got only the vote's pages and answered in JSON with the label and up to three sentences copied from the booklet, which we looked up in the PDF, exactly or as the closest passage, to give the evidence with its real page, after answers backed by verified quotes [12]. It reached 0.850 and a Hit@5 of 0.485, for 11,553 input tokens per case. Asking for quotes on neutral answers too raised Hit@5 to 0.642 but lowered macro-F1 to 0.836. Retrieving the sentences ourselves cut the tokens about tenfold and makes the evidence verbatim by construction.
- **Paraphrases** of the claim in the search [10], merged by reciprocal rank fusion [5]: 0.979 in the beginner task, and 0.947 instead of 0.951 in the advanced task, for three times and 1.25 times the tokens. They turned more neutral claims into contradictions, mostly claims about another vote whose explanation said the booklet does not mention them. Reading such answers as neutral removed nearly all of these, but also some true contradictions explained the same way ("the booklet does not mention five years, it says ten").
- **Also searching the booklet's edition in the claim's language,** each sentence found mapped to the same position on the same page [8]: 0.962 instead of 0.951, left out for the reason in Section 2.3.
- **Less context in the beginner task,** 1,500 or 1,000 characters: 0.984 and 0.977, for 800 and 729 tokens per case instead of 834.
- **Four prompt rules** against the main advanced errors: 0.973, for 11 % more tokens. They fixed 15 errors and made 17 new ones.

Tokens and time. Narrowing the context cut the tokens per case 2.3-fold in the beginner task and 30-fold in the advanced task, while it raised macro-F1. Without Apertus' latency (1.5 s and 1.9 s per case), a run of all 1,488 cases takes 0.9 s in the beginner task, all of it start-up, and 5.4 s in the advanced task, 4.6 s of it reading

the booklets' text layers before the first request. Their tags are read while requests are in flight.

6 Limitations

- **Prompt engineering only:** besides the paraphrasing approach did not test more complicated methods.
- **Errors left:** most are wrong contradictions in the advanced task: 18 neutral claims, and 10 entailments, mostly same-language claims about what happens if the vote is accepted. In the beginner task, scoping cut away the deciding sentence of 5 contradictions.
- **Booklet structure:** narrowing to the vote relies on running headers. A booklet without them is searched whole. The booklet without tags is read without subheadings; 14 of its 17 cases are right, and the 3 wrong are one repeated claim.
- **No OCR:** a scanned booklet would have neither tags nor a text layer.
- **Time** is measured in-process on a laptop. The organisers time the whole container, including its start.
- **Task difficulty:** the task would be closer to reality, if the correct booklet was not given, but had to be found by Apertus itself; especially considering that some voting questions appear again over time (e.g. about Switzerland joining the EU), this would increase the task difficulty potentially by quite a lot.

7 Reproducibility

- **Setup:** `uv sync` in `track_2a/`, then `uv run python -m src stage=prepare` for the dataset and the booklets. Dependencies (`uv.lock`) and the dataset revision are pinned.
- **Runs:** `uv run python -m src strategy=gied_beginner_scoped model=apertus-70b` and `uv run python -m src task=advanced strategy=gied_strict_reasoned model=apertus-70b`, with their results in `data/runs/`.
- **Submission:** `make run` builds the Docker image and decides the sample cases. It needs `API_KEY` (and `BASE_URL` for another endpoint), any CPU, no GPU and no model weights.

8 Next steps

- **The search, more than the prompt.** With the source given, the task is largely one of prompt refinement. The label definitions and reasoning first alone raised beginner macro-F1 from 0.788 to 0.962 (Table 2). Fact verification proper starts by finding the source [18]: which of the 60 booklets, which edition and which vote. That is not trivial, since topics such as immigration or the relationship with the EU come back to the vote over the years.
- **Above 0.9, the labels are open to interpretation.** On MNLI, the majority of five annotators matches the gold label in only about 92 % of the cases [13], and such disagreement is often genuine [15]. Our runs show it too: four extra prompt rules fixed 15 errors and made 17 new ones. Several people relabelling our errors would tell mistakes from disagreements.
- **Smaller steps:** test the cheaper 8B model, and read each booklet's text layer only when a case first needs it.

9 License

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References

[1] Mikel Artetxe, Vedanuj Goswami, Shruti Bhosale, Angela Fan, and Luke Zettlemoyer. Revisiting machine translation for cross-lingual classification. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6489–6499, Singapore, December 2023. Association for Computational Linguistics.

[2] Christoph Auer, Maksym Lysak, Ahmed Nassar, Michele Dolfi, Nikolaos Livathinos, Panos Vagenas, Cesar Berrospi Ramis, Matteo Omenetti, Fabian Lindlbauer, Kasper Dinkla, Valery Weber, Lucas Morin, Ingmar Meijer, Viktor Kuropiatnyk, and Peter Staar. Docling technical report, 08 2024.

[3] Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, Xinran Zhao, Hongming Zhang, and Dong Yu. Dense X retrieval: What retrieval granularity should we use? In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15159–15177, Miami, Florida, USA, November 2024. Association for Computational Linguistics.

[4] Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel R. Bowman, Holger Schwenk, and Veselin Stoyanov. XNLI: Evaluating cross-lingual sentence representations. In Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2475–2485, Brussels, Belgium, October-November 2018. Association for Computational Linguistics.

[5] Gordon V. Cormack, Charles L A Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '09, page 758–759, New York, NY, USA, 2009. Association for Computing Machinery.

[6] Eric Egli, Noah Mamié, Eyal Dolev, and Mathias Müller. Voting booklet bias: Stance detection in swiss federal communication, 06 2023.

[7] Julen Etxaniz, Gorka Azkune, Aitor Soroa, Oier Lopez de Lacalle, and Mikel Artetxe. Do multilingual language models think better in English? In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 550–564, Mexico City, Mexico, June 2024. Association for Computational Linguistics.

[8] William A. Gale and Kenneth W. Church. A program for aligning sentences in bilingual corpora. *Computational Linguistics*, 19(1):75–102, 1993.

[9] Alejandro Hernández-Cano, Alexander Hägele, Allen Hao Huang, Angelika Romanou, Antoni-Joan Solergibert, Barna Pásztor, Bettina Messmer, Dhia Garbaya, Eduard Frank Ďurech, Ido Hakimi, Juan Garcia Giraldo, Mete Ismayilzada, Negar Foroutan, Skander Moalla, Tiancheng Chen, Vinko Sabolčec, Yixuan Xu, Michael Aerni, Badr AlKhamissi, Inés Altemir Marinas, Mohammad Hossein Amani, Matin Ansaripour, Ilia Badanin, Harold Benoit, Emanuela Boros, Nicholas John Browning, Fabian Bösch, Maximilian Böther, Niklas Canova, Camille Challier, Clément Charmillot, Jonathan Coles, Jan Milan Deriu, Arnout Devos, Lukas Drescher, Daniil Dzenhaliou, Maud Ehrmann, Dongyang Fan, Simin Fan, Silin Gao, Miguel Gila, María Grandury, Diba Hashemi, Alexander Miserlis Hoyle, Jiaming Jiang, Mark Klein, Andrei Kucharavy, Anastasiia Kucherenko, Frederike Lübeck, Roman Machacek, Theofilos Ioannis Manitaras, Andreas Marfurt, Kyle Matoba, Simon Matrenok, Henrique Mendonça, Fawzi Roberto Mohamed, Syrielle Montariol, Luca Mouchel, Sven Najem-Meyer, Jingwei Ni, Gennaro Oliva, Matteo Pagliardini, Elia Palme, Andrei Panferov, Léo Paoletti, Marco Passerini, Ivan Pavlov, Auguste Poiroux, Kaustubh Ponkshe, Nathan Ranchin, Javier Rando, Mathieu Sauser, Jakhongir Saydaliev, Mukhammadali Sayfiddinov, Marian Schneider, Stefano Schuppli, Marco Scialanga, Andrei Semenov, Kumar Shridhar, Raghav Singhal, Anna Sotnikova, Alexander Sternfeld, Ayush Kumar Tarun, Paul Teiletche, Jannis Vamvas, Xiaozhe Yao, Hao Zhao, Alexander Ilic, Ana Klimovic, Andreas Krause, Caglar Gulcehre, David Rosenthal, Elliott Ash, Florian Tramèr, Joost VandeVondele, Livio Veraldi, Martin Rajman, Thomas C. Schulthess, Torsten Hoefler, Antoine Bosselut, Martin Jaggi, and Imanol Schlag. Apertus: Democratizing open and compliant LLMs for global language environments. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens, editors, *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 46877–46955, San Diego, California, United States, July 2026. Association for Computational Linguistics.

[10] Alexander Hoyle, Rupak Sarkar, Pranav Goel, and Philip Resnik. Natural language decompositions of implicit content enable better text representations. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13188–13214, Singapore, December 2023. Association for Computational Linguistics.

[11] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

[12] Jacob Menick, Maja Trębacz, Vladimir Mikulik, John Aslanides, Francis Song, Martin Chadwick, Mia Glaese, Susannah Young, Lucy Campbell-Gillingham, Geoffrey Irving, and Nat McAleese. Teaching language models to support answers with verified quotes, 03 2022.

[13] Nikita Nangia and Samuel R. Bowman. Human vs. Muppet: A conservative estimate of human performance on the GLUE benchmark. In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4566–4575, Florence, Italy, July 2019. Association for Computational Linguistics.

[14] OST. MNLI over Swiss voting booklets: Benchmark dataset. Hugging Face dataset, `https://huggingface.co/datasets/OSTswiss/MNLIoverSwissVotingBooklets`, October 2026. Version v1.1, revision `fc2b276`. AI Applications and Deployment Lab, Institute for Software, Eastern Switzerland University of Applied Sciences OST.

[15] Ellie Pavlick and Tom Kwiatkowski. Inherent disagreements in human textual inferences. *Transactions of the Association for Computational Linguistics*, 7:677–694, 2019.

[16] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends® in Information Retrieval*, 4:1–174, 09 2009.

[17] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023.

[18] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In Marilyn Walker, Heng Ji, and Amanda Stent, editors, *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, New Orleans, Louisiana, June 2018. Association for Computational Linguistics.

[19] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA, 2022. Curran Associates Inc.

[20] Peng Xu, Wei Ping, Xianchao Wu, Lawrence McAfee, Chen Zhu, Zihan Liu, Sandeep Subramanian, Evelina Bakhturina, Mohammad Shoeybi, and Bryan Catanzaro. Retrieval meets long context large language models. In *The Twelfth International Conference on Learning Representations*, 2024.